

MTH 377/577 Convex Optimization

Problem set 1

1. Is the set of vectors $\{(a, b) \in R^2 : b = 3a + 1\}$ a vector space?
2. Is the set of all $x \in R^3$ such that $\|x\| = 1$ a subspace of R^3 ?
3. Are the following vectors linearly independent?
 - (a) $v_1 = (2, 6, 3), v_2 = (1, -1, 4), v_3 = (3, 2, 1), v_4 = (2, 5, 4)$
 - (b) $v_1 = (1, -1, 3, -1), v_2 = (1, -1, 4, 2), v_3 = (1, -1, 5, 7)$
4. Find a basis for the subspace of R^n spanned by the following vectors:
$$\{(1, 3, 3), (-3, -9, -9), (1, 5, -1), (2, 7, 4), (1, 4, 1)\}$$
5. Is the set of possible distances between some point on the circle $x^2 + y^2 = 1$ and any point on the circle $x^2 + y^2 = 100^2$ bounded?
6. Are the following sets open or closed?
 - (a) The empty set ϕ
 - (b) The complement of $\{x = 5\}$ in R .
7. Suppose f is a real valued continuous function and its domain $D \subset R$ is convex at all points except at $x = 1$. Does f have a maxima or minima?
8. Is the set of points $\{x^k\}_{k=1,2,\dots}$ where $x^k = k^{\frac{1}{2}}$ for each $k = 1, 2, \dots$ a closed set?
9. Every affine set is convex. True/False?
10. Let $C_n = \{(x_1, \dots, x_n) \in R^n : \sum_{i=1,2,\dots,n} \alpha x_i \leq n; \forall \alpha \in I_+\}$. Is $C_n \cup C_{n+1}$ for some n a closed set? Is it bounded?
11. Let $A = \{(x, y) \in R^2 | x^2 + y^2 < 4\}$ and $B = \{(x, y) \in R^2 | x + y = 4\}$. For any $(x, y) \in B$, can you find a hyperplane that separates it from A ?